PPBv2 localization architecture (VIO, local fusion, global fusion)

Legend
built, in repo
to build, Stage 1A / 1B
future, Layer 3 / live
hardware prerequisite
safety, untouched
existing PPBv2
ROS topic

Sensors
Trigger board
Arduino strobe
trigger
trigger
sync
Stereo camera left
planned global-shutter nav camera
Stereo camera right
planned global-shutter nav camera
Raw IMU ≥ ±16 g, 100-200 Hz
hardware-synced to trigger
NEW HARDWARE, Task 5
HWT905 is AHRS-only
Amiga CAN / gRPC
wheel speeds
UM982 dual-antenna RTK
ANT1 position, ANT2 heading
NTRIP RTCM3 corrections

Offline verification
Datasets, CPU, no robot
EuRoC, Rosario v1
KITTI, KAIST
run_bench.sh + stress suite
frame drop, jitter, blackout,
IMU drop, mid-row start
offline serial runner
GPS dropout / degradation injector
degraded /gps/fix
validation only, never reaches motors before Task 8
Scoring + acceptance gate
ATE, RTE, NEES, yaw drift, coverage
jump-free re-anchor

Calibration · Task 5
Kalibr
stereo + camera-IMU
spatial + temporal
Allan variance
IMU noise parameters
kalibr_to_openvins.py
versioned calibration folder
+ platform.yaml health limits
left image
right image
estimator config

Drivers
IMU driver
hardware timestamps on Orin
wheel_odom_node, Task 6
on Pi 5
calibrated scale + track width
um982_driver
exists, on Pi 5
Static TF publisher
base_link → imu_link, cam_left, cam_right, gps_ant1, gps_ant2
/imu/data
sensor_msgs/Imu, imu_link
dual-antenna heading
same inputs as ekf_local
absolute yaw when valid
TODAY: RTK pose as Odometry, map → base_link
retired at Task 8
TODAY: rtk_status_flag
publishes TF

Layer 1 · VIO
OpenVINS
Jetson AGX Orin, ROS 2 Humble, CPU
MSCKF stereo-inertial
ZUPT, online camera-IMU calibration
publishes NO TF
raw OpenVINS odometry
vio_odom_adapter
IMU → base_link extrinsic
covariance_scale
vio_health divergence gate
/vio/odom
nav_msgs/Odometry + covariance
vio_world → base_link
/vio/status
DiagnosticArray
OK / WARN / DIVERGED

Layer 2 · local fusion
vio_gate, NEW
withhold while DIVERGED
on recovery: drop first message or re-base
so the differential delta never jumps
/vio/odom_gated
ekf_local,
robot_localization
CPU, on Orin or Pi 5
world_frame = odom, 50 Hz, two_d_mode
imu0: gyro + accel
odom0: wheel vx, vyaw
odom1: VIO pose, DIFFERENTIAL
/odometry/local
odom → base_link
publishes TF

Layer 3 · global fusion · FUTURE
/wheel/odom
nav_msgs/Odometry
vx, yaw rate, vy = 0
/gps/fix
sensor_msgs/NavSatFix
gps_quality_gate
Task 4 offline, Task 8 live
covariance from fix type, diff-age,
HDOP/VDOP, heading σ
tiny when fixed, inflated when float,
effectively infinite when denied
GPS quality
/gps/fix_gated
health_supervisor
healthy → degraded → denied → recovery
ekf_global,
robot_localization
world_frame = map
imu0, odom0, odom1 as in ekf_local
odom2: GPS absolute x, y
yaw: dual-antenna heading when valid
publishes TF
/odometry/global
map → base_link
/gps/datum
degradation state
odometry yaw
Task 8: repoint follower
covariance bound
navsat_transform_node
30 Hz, zero_altitude,
use_odometry_yaw
wait_for_datum, datum = /gps/datum
/odometry/gps
map frame

Navigation + safety
waypoint_follower
Raspberry Pi 5, ROS 2 Jazzy
rtk_monitor
hard stop kept as a covariance-bounded backstop
NEVER removed
velocity command
stop
twist_mux
Feather M4 CAN bridge
Amiga motors

TF tree · one owner each
map
ENU, transverse Mercator at datum
may jump
ekf_global, rate-limited
odom
continuous, smooth
ekf_local
base_link
static
imu_link, cam_left, cam_right, gps_ant1, gps_ant2